Songying Technology Co., Ltd.

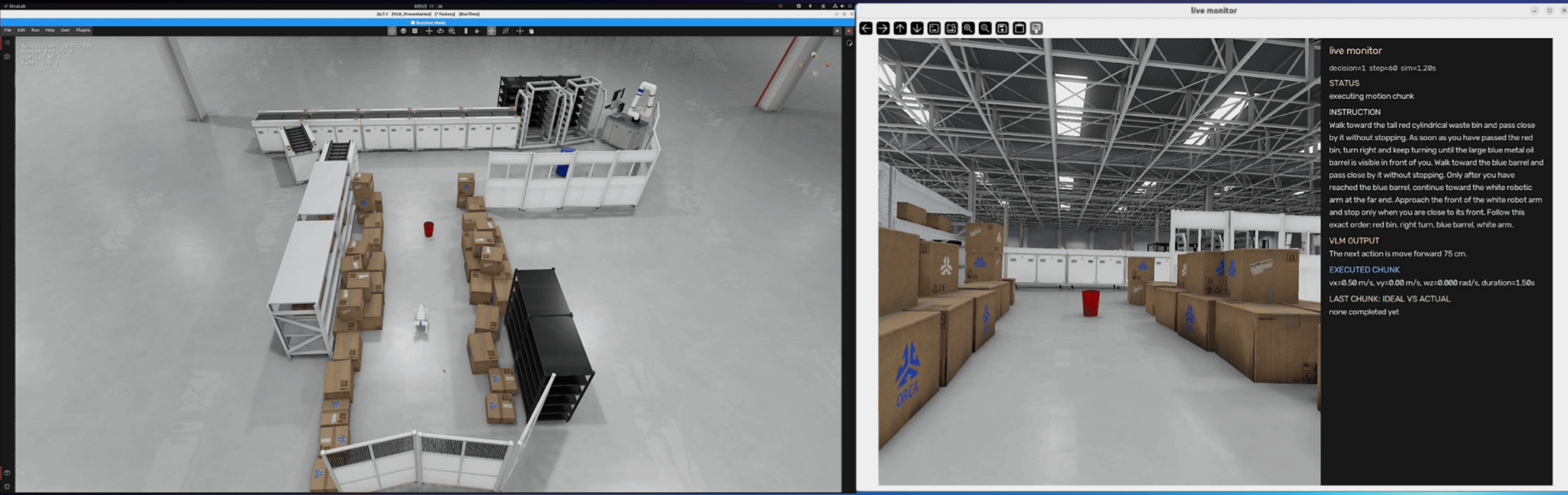
松应科技 ORCA

ORCA VLN
Hackathon Training Manual

Reproduce Visual-Language Navigation in OrcaLab

Simulation First · Visible Loop · Verifiable Improvements

Ego RGB → NaVILA → Velocity Command → Orca Lab


live monitor
live monitor
decision=1 step=60 sim=1.20s
STATUS
executing motion chunk
INSTRUCTION
Walk toward the tall red cylindrical waste bin and pass close by it without stopping. As soon as you have passed the red bin, turn right and keep turning until the large blue metal oil barrel is visible in front of you. Walk toward the blue barrel and pass close by it without stopping. Only after you have reached the blue barrel, continue toward the white robotic arm at the far end. Approach the front of the white robot arm and stop only when you are close to its front. Follow this exact order: red bin, right turn, blue barrel, white arm.
VLM OUTPUT
The next action is move forward 75 cm.
EXECUTED CHUNK
vx=0.50 m/s, vy=0.00 m/s, wz=0.000 rad/s, duration=1.50s
LAST CHUNK: IDEAL VS ACTUAL
none completed yet
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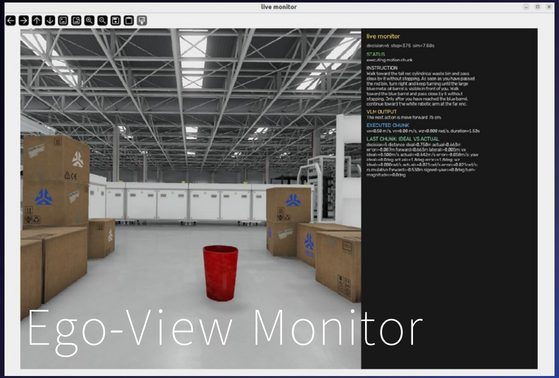
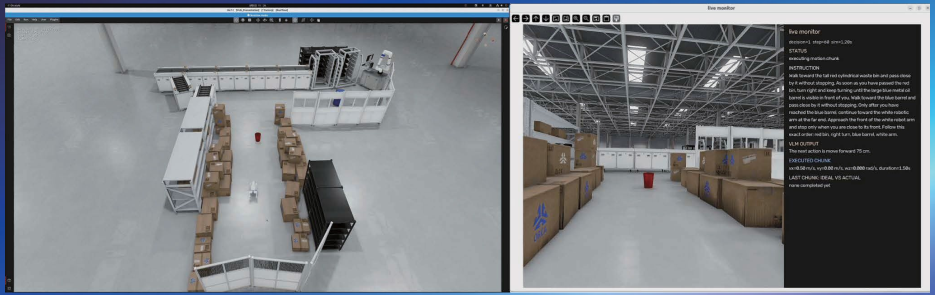
What You Will Build and Show

1. Instruction + 8 Ego RGB Frames → NaVILA → One Navigation Action
2. A Velocity Chunk Drives the Robot; the Scene Returns the Next Observation Save
3. RGB, Action / State Trajectories, and measurements.json

Instruction+ RGB History → NaVILA Baseline [1] http://navila-bot.github.io/ → Navigation Action → Orca Lab Returns the Next Frame

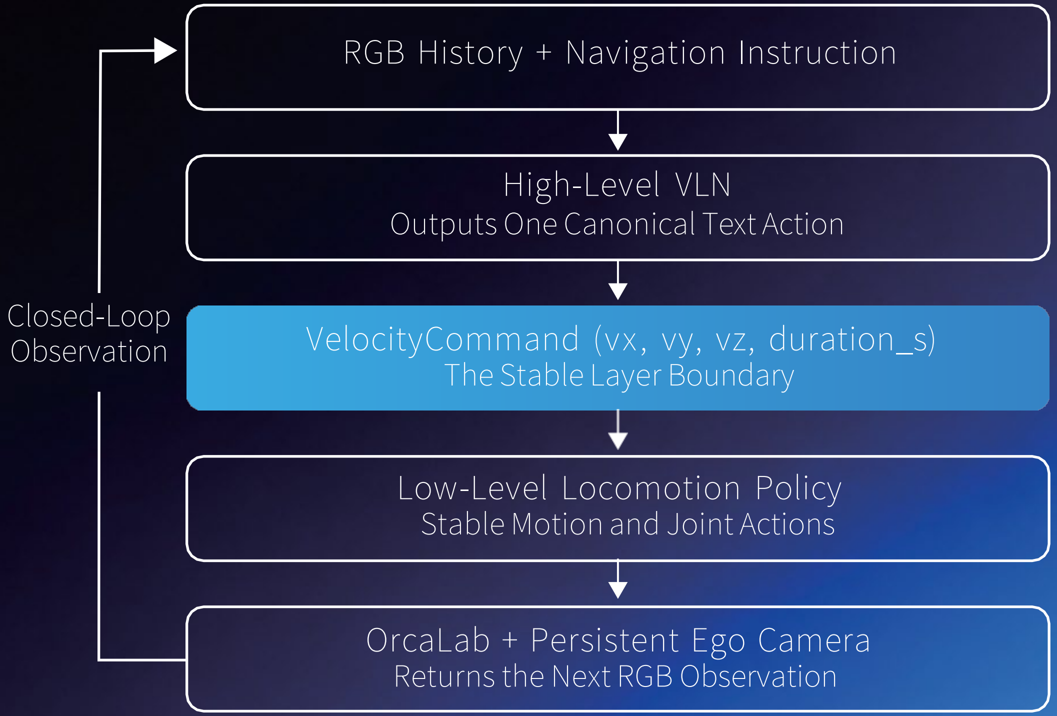




[1] Cheng et al., NaVILA: Legged Robot Vision-Language-Action Model for Navigation, RSS 2025.

The Shared Boundary: Improve One Layer Without Breaking the Other

Teams can improve the decision model or the motion policy independently.



Above the Boundary

- Prompt and Route Logic
- SFT / LoRA
- Inspection Logic

Below the Boundary

- Gait and Recovery
- Command Tracking
- New Policy Adapter

Keep the Velocity Command Semantics Unchanged

Separate Terminal For Testing

Run them separately to locate problems quickly.

A

Orca Lab

Open IndustrialWarehouse1_3dgs and import default_set.json

```
./scripts/start_orcalab_gui.sh
```

B

NaVILA

Server Listening 127.0.0.1:54321

```
./scripts/start_navilm_server.sh
```

C

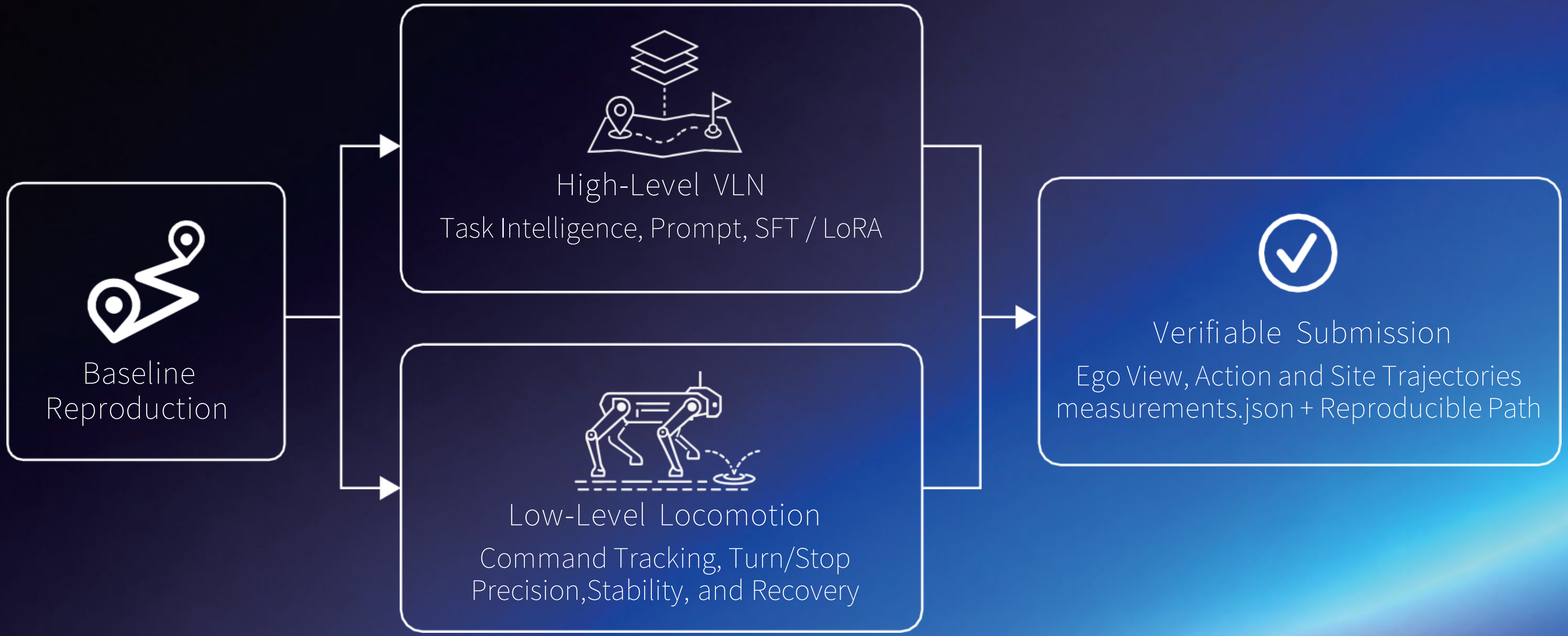
Run the Baseline

Run Closed-Loop Navigation and Record Evidence

```
./scripts/run_orcalab_scene_locomotion.sh
```

Check First: One Complete Go2 Actor • Correct Version • Valid Model Path

Choose Your Improvement Track



High-Level VLN Adaptation: Use SFT / LoRA Carefully

1

Run Rollouts

- instruction
- image_paths
- baseline_actions

2

Human Review

Align Images with Canonical Actions

3

SFT / LoRA

Change Only the High-Level VLN

4

Fixed-Episode Evaluation

- Valid Actions
- Visual Grounding
- Closed-Loop Outcome

✓ Fix the Low-Level Policy; Save and Review Rollouts First

✓ A Review Queue Is Not Ground Truth: Label Only After Image–Action Alignment

✓ Use Fixed Scenes for Regression: Valid Actions, Visual Grounding, and Closed-Loop Outcome

Low-Level Locomotion: Train Anywhere, Integrate Precisely

Compatible Checkpoint

Use a Platform-Native Low-Level Policy; Load and Run Directly

Backend Adapter

Use a Backend Adapter to Bridge the Chosen Platform

VelocityCommand (vx, vy, wz, duration_s)

Keep the Velocity Command Interface Unchanged

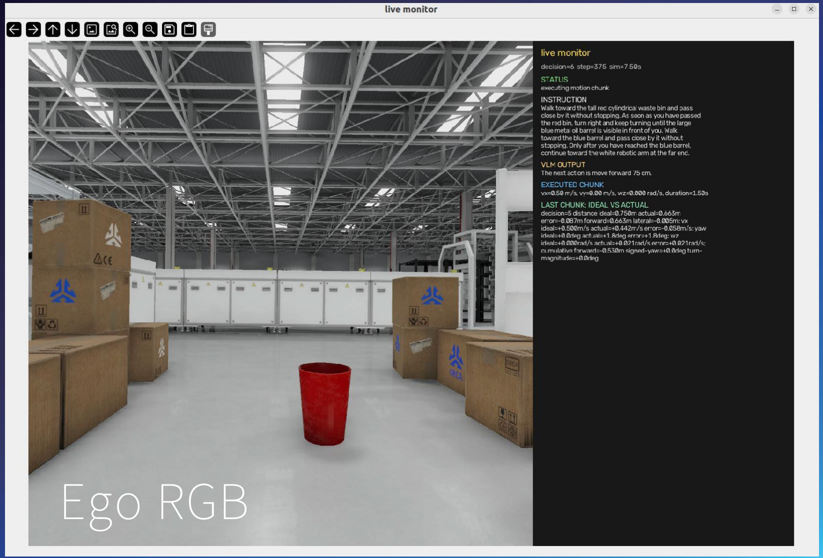
Alignment Checklist (Before Integration)

- ✓ Robot Asset
- ✓ Joint / Observation ABI
- ✓ Control Timing
- ✓ State Bridge
- ✓ Stand, Forward, Turn
- ✓ Stop and Recovery

- Train in OrcaLocomotion, IsaacLab, or Another Platform
- Replace Directly When ABI-Compatible; Otherwise Write an Adapter
- Test Basic Motions Before Connecting the VLM

Night 1

Install all the environment and open orcalab, import the scene.
Try run the demo!

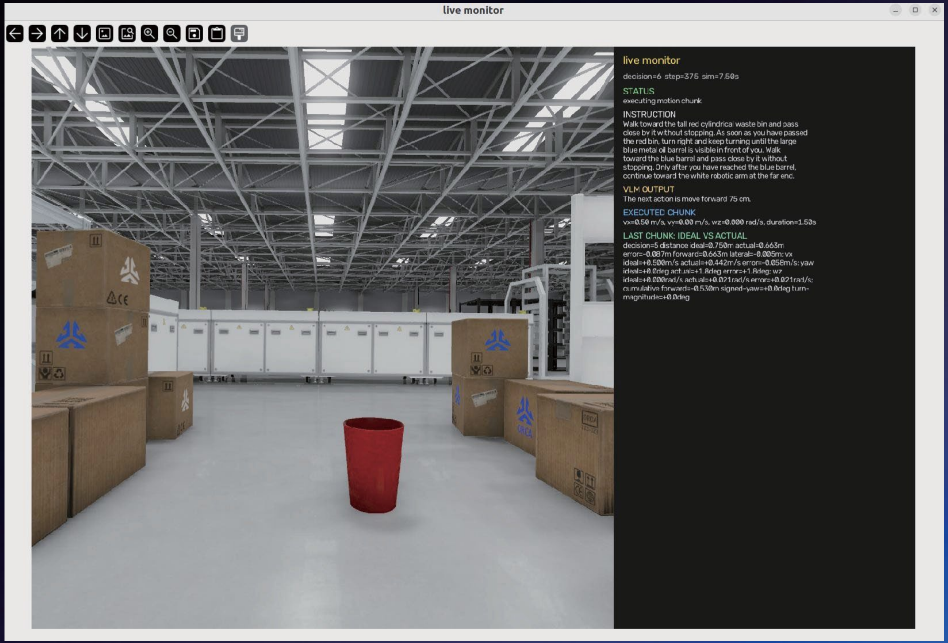


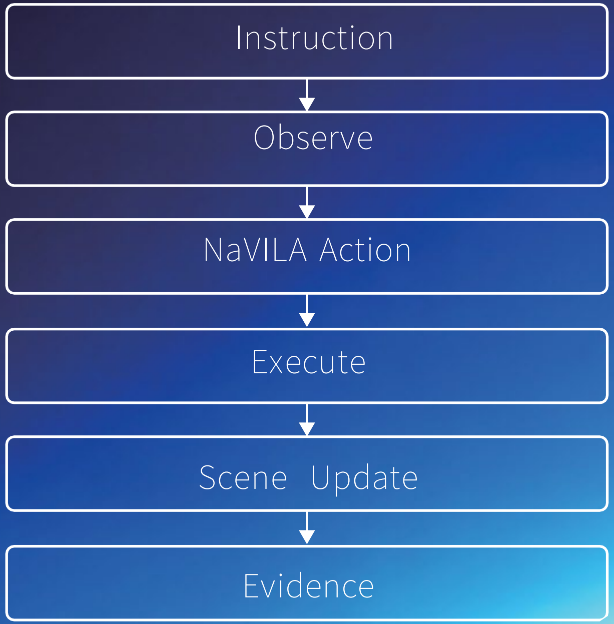
Create Once → Update Pose → Get PNG

- ☑ Exactly One Complete Go2 in the Factory Scene
- ☑ Create an ego camera and make sure its working
- ☑ Warm up for 100 steps to adjust start position

Night 2

Run and Read a Closed-Loop Navigation Episode





Default task: Approach red waste bin, large blue oil barrel, white robotic arm separately.
Model returns one action chunk at one time.
Try to avoid surrounding objects.

Night 2

Improve the Baseline Without Losing Evidence

1
Baseline Reproduction
Compare Two Action Sequences and Final Trajectories

2
Language Ablation
Change Only the Instruction Wording

3
Camera Ablation
Change Only the Camera Pose; Compare RGB and Actions

Instruction → Camera → Action → Path

Change One Variable at a Time

Evidence Table	Ego RGB	Action Sequence	State Trajectory	measurements.json

Make the Loop Visible and Reproducible First; Then Change One Layer at a Time.

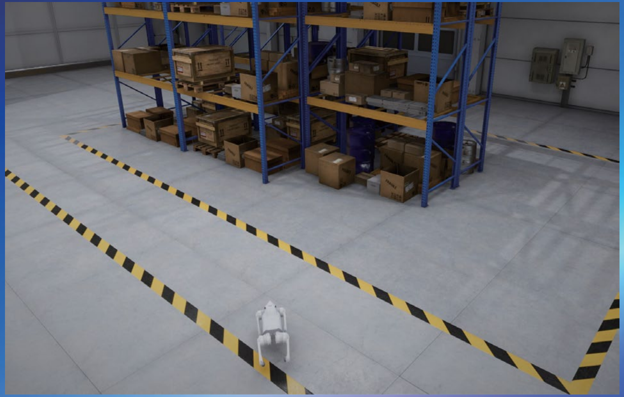
Night 2

Improve the Baseline Without Losing Evidence

- **Run Path**
  Reproduce with One Command
- **Scene and Configuration**
  Save Configuration Notes
- **Run Evidence**
  View Images, Video, and Logs
- **Submission Notes**
  Improvement Track and Baseline Comparison

Evidence Package

- Ego-View Images or Video
- measurements.json and Logs
- Replayable Run Path



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Thanks for Watching

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